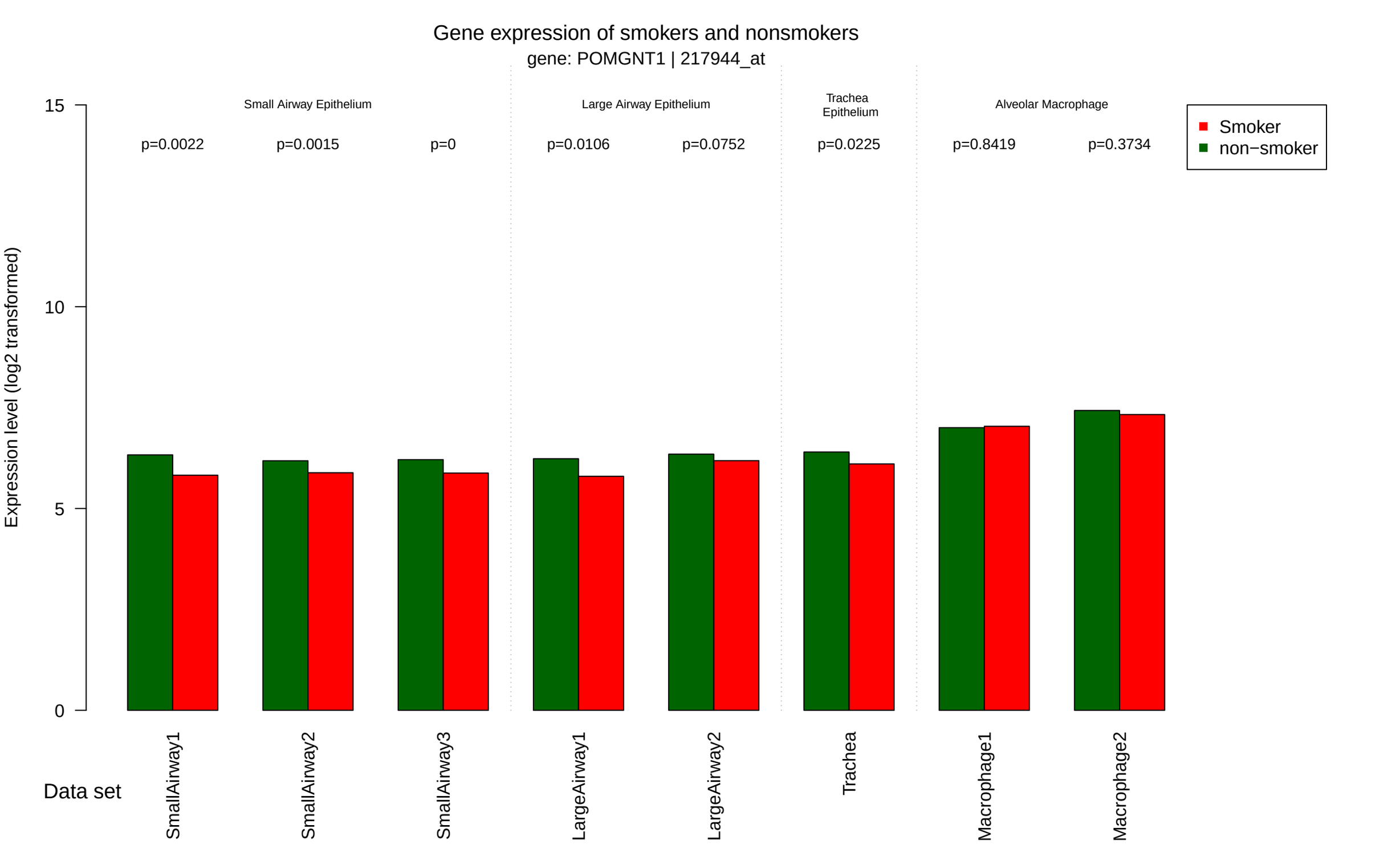
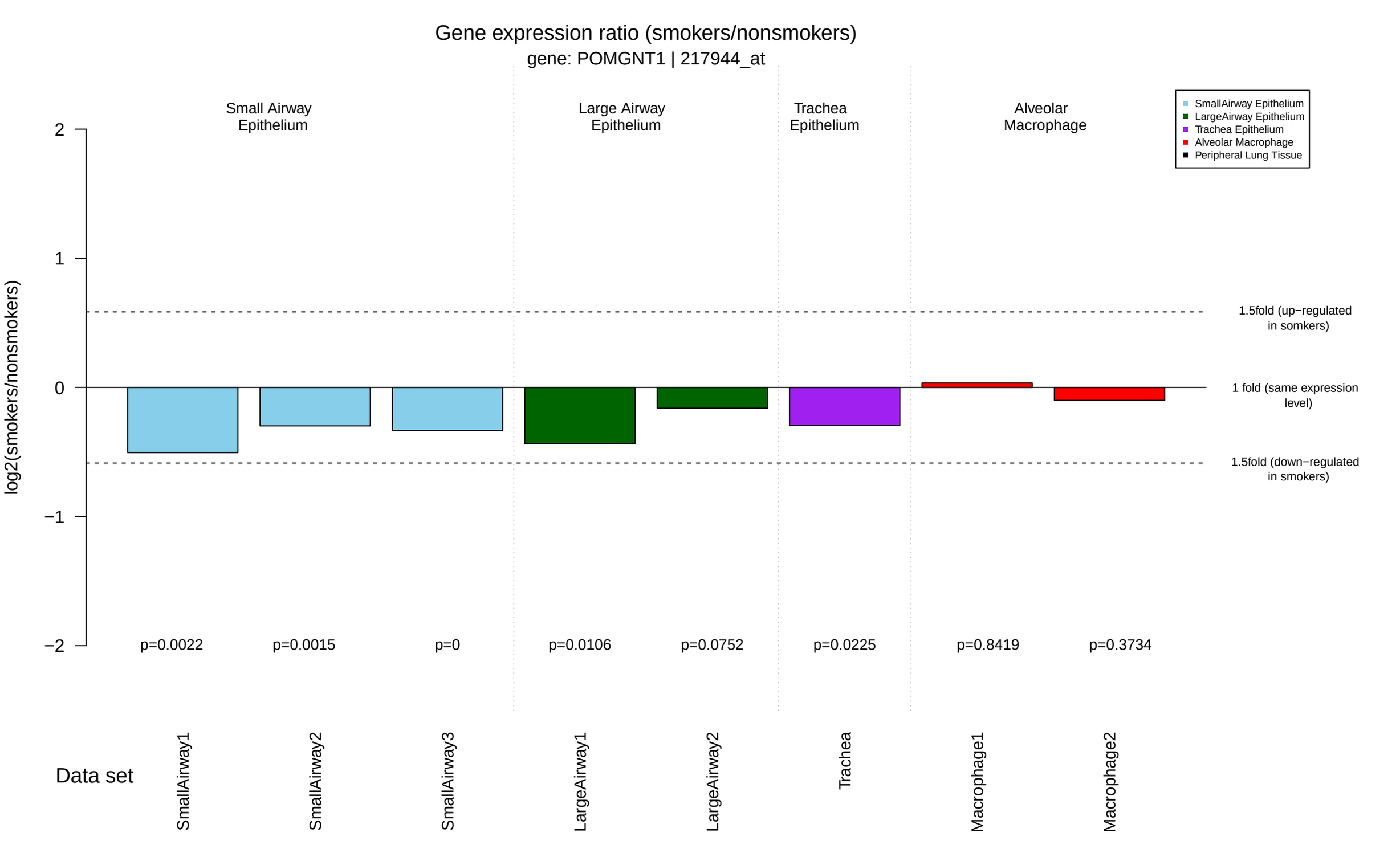
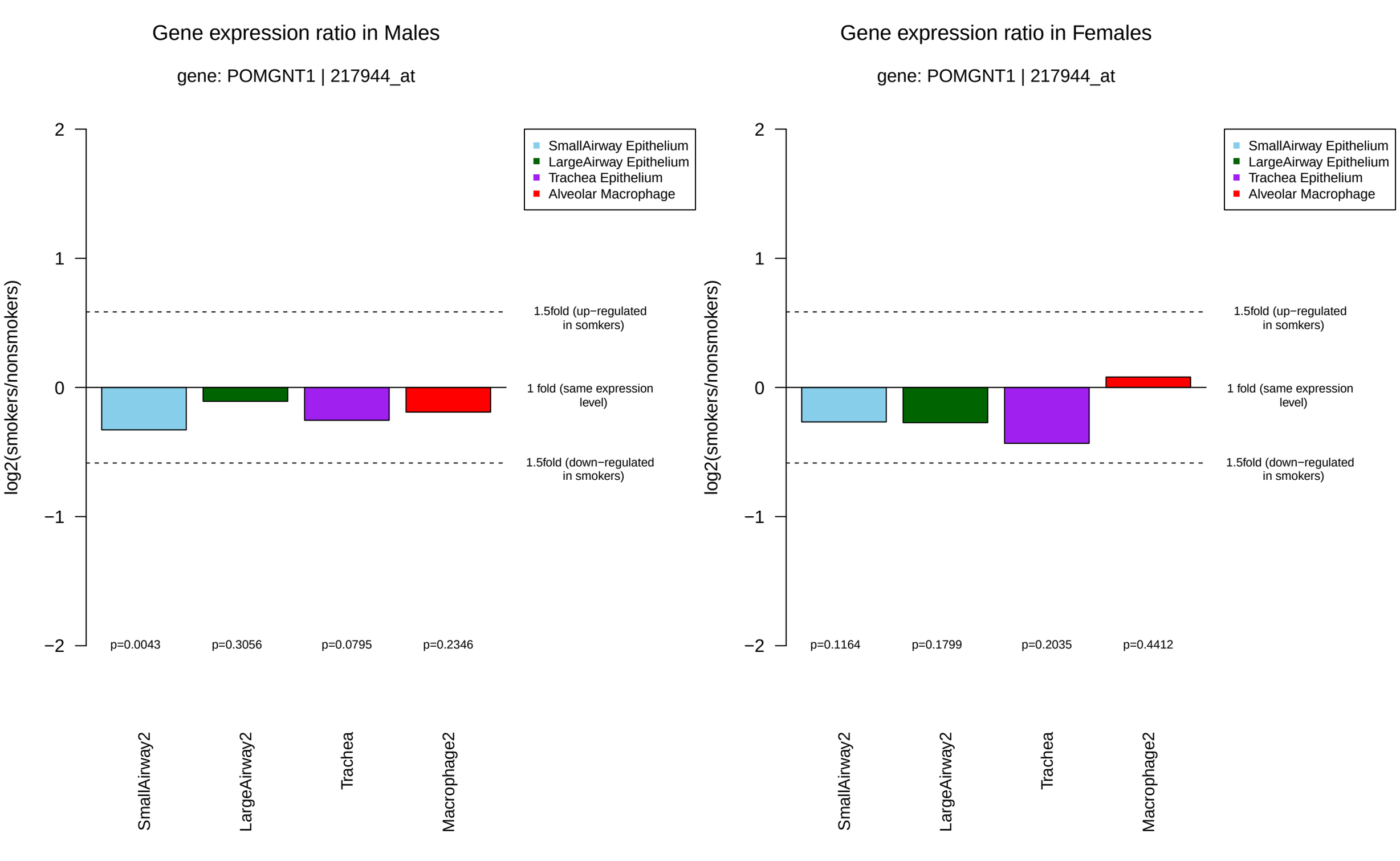
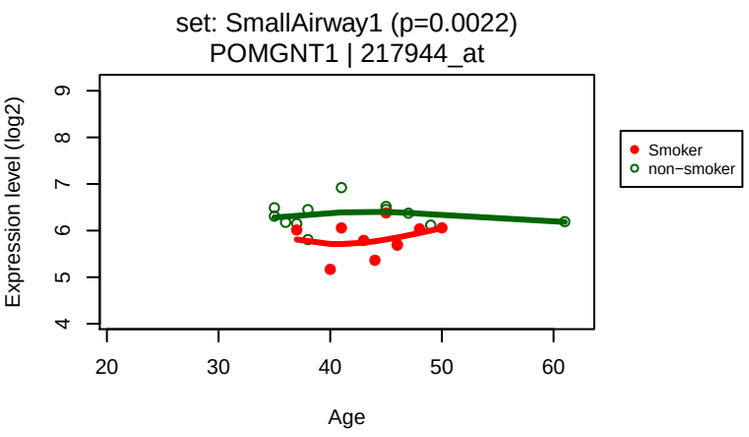
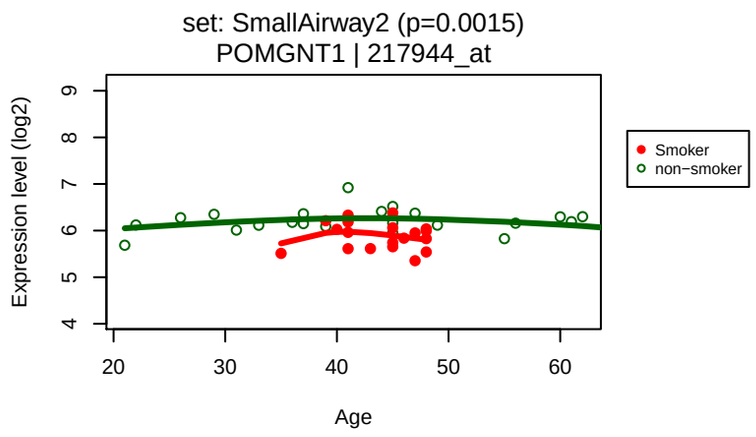
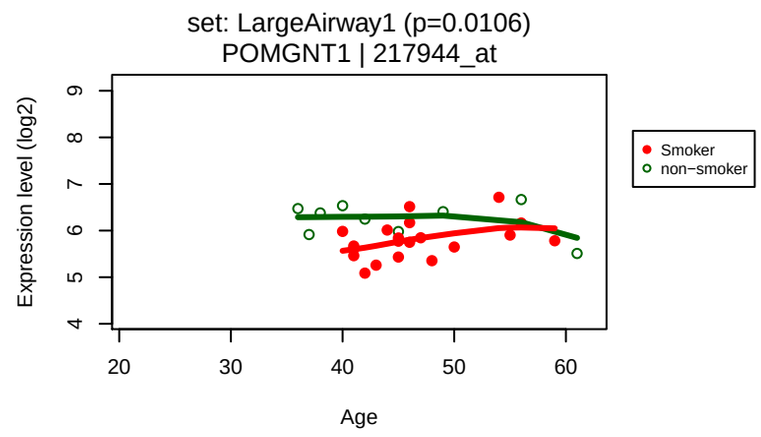

Gene expression of smokers and nonsmokers
gene: POMGNT1 | 217944_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0022
p=0.0015
p=0
p=0.0106
p=0.0752
p=0.0225
p=0.8419
p=0.3734
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

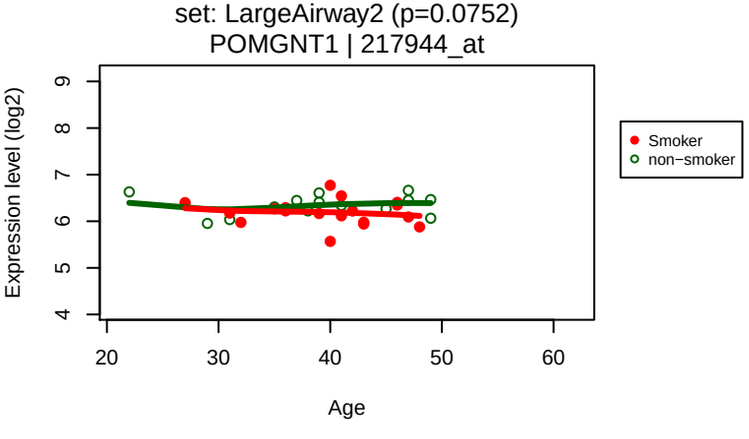

Gene expression ratio (smokers/nonsmokers)
gene: POMGNT1 | 217944_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0022
p=0.0015
p=0
p=0.0106
p=0.0752
p=0.0225
p=0.8419
p=0.3734
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

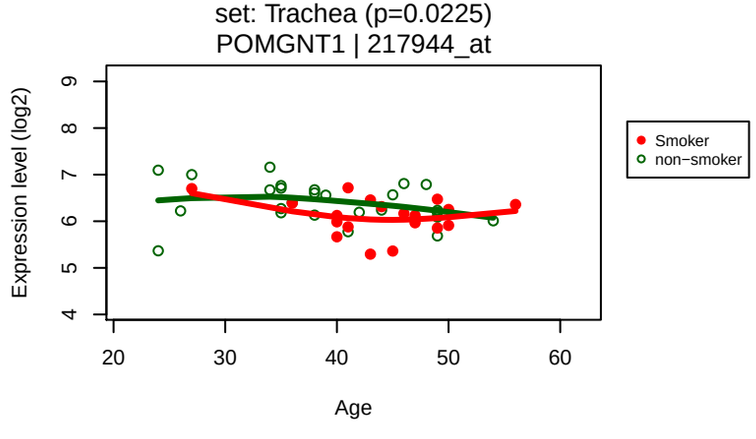

Gene expression ratio in Males
gene: POMGNT1 | 217944_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0043
p=0.3056
p=0.0795
p=0.2346
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: POMGNT1 | 217944_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1164
p=0.1799
p=0.2035
p=0.4412
SmallAirway2
LargeAirway2
Trachea
Macrophage2

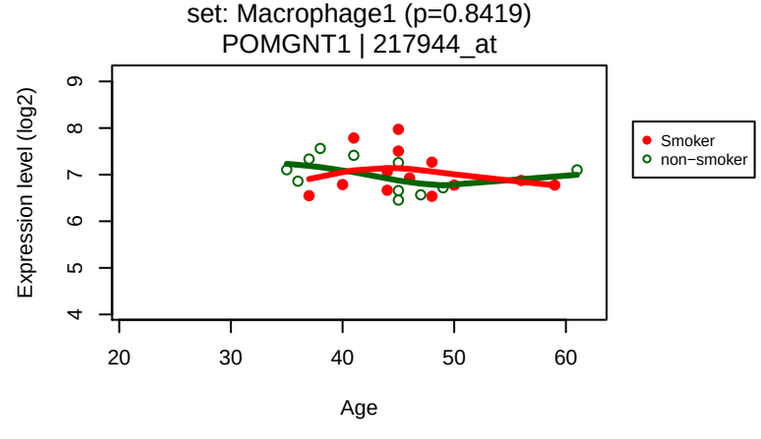

set: SmallAirway1 (p=0.0022)
POMGNT1 | 217944_at
Expression level (log2)
Age
Smoker
non−smoker

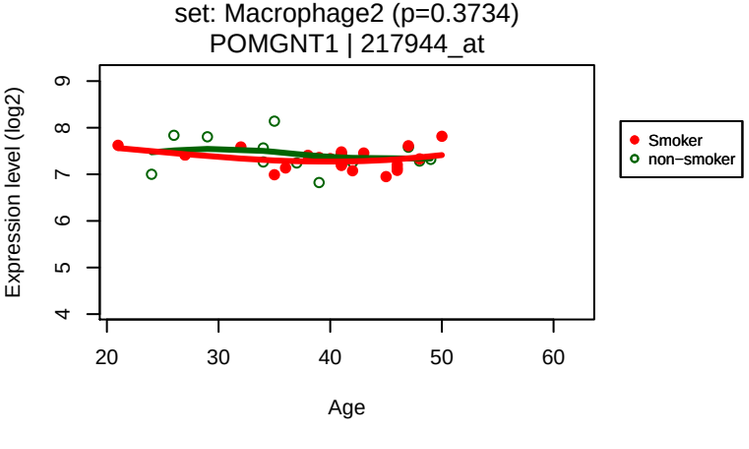

set: SmallAirway2 (p=0.0015)
POMGNT1 | 217944_at
Expression level (log2)
Age
Smoker
non−smoker

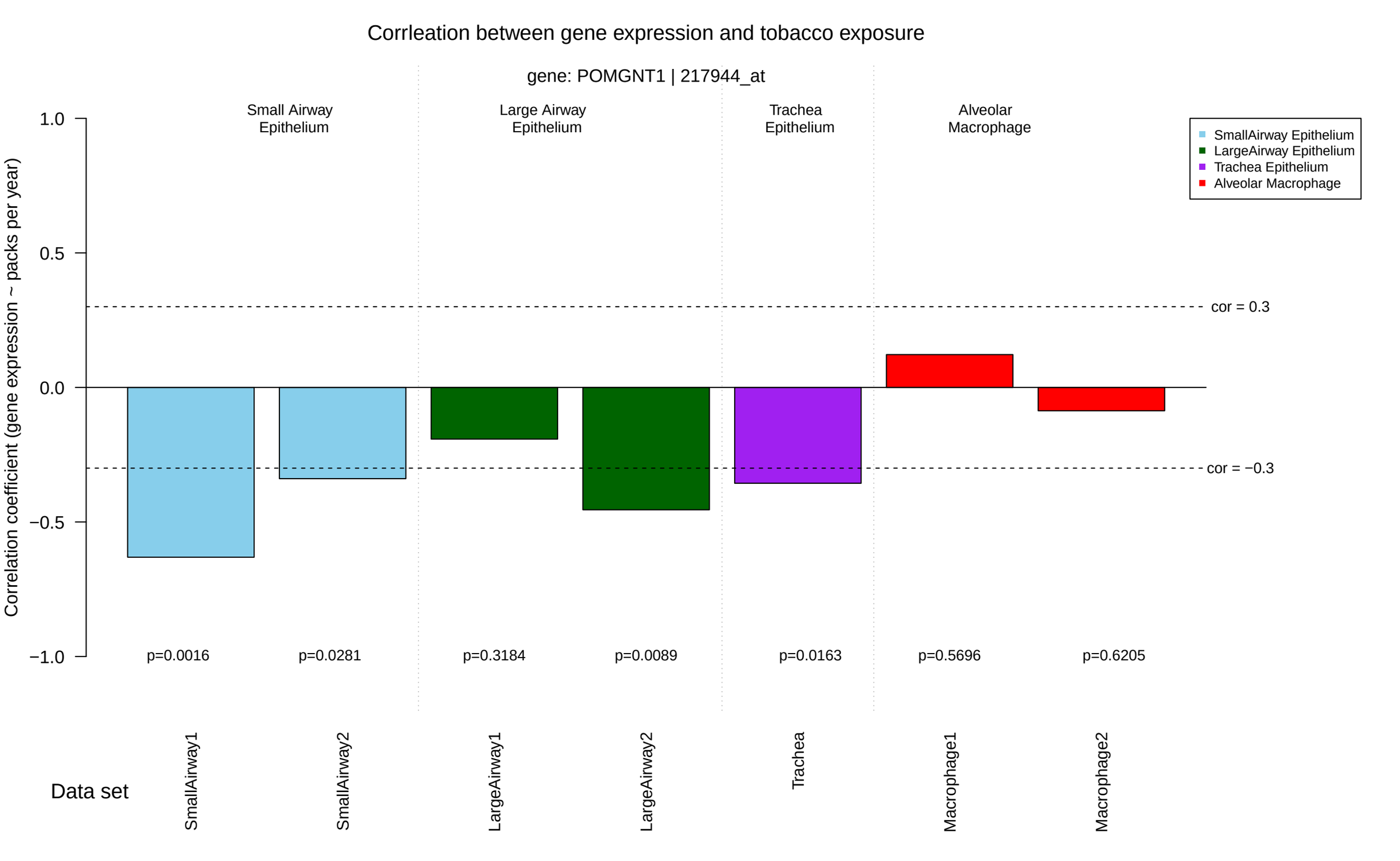

set: LargeAirway1 (p=0.0106)
POMGNT1 | 217944_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0752)
POMGNT1 | 217944_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0225)
POMGNT1 | 217944_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.8419)
POMGNT1 | 217944_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3734)
POMGNT1 | 217944_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: POMGNT1 | 217944_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0016
p=0.0281
p=0.3184
p=0.0089
p=0.0163
p=0.5696
p=0.6205
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2